

**11.16** Solution: (a) In the rest frame of the medium, we have  $\mathbf{J} = \sigma \mathbf{E}$ . Using the field strength tensor, we can write it as

$$J^i = \sigma F^{i0} = \frac{\sigma}{c} F^{i\beta} U_\beta,$$

where  $U^\beta = (c, \mathbf{0})$  in the rest frame. It is natural to extend this result to the covariant form, with

$$J^\alpha = \frac{\sigma}{c} F^{\alpha\beta} U_\beta.$$

However, due to the antisymmetry of  $F^{\alpha\beta}$ ,

$$U_\alpha J^\alpha = \frac{\sigma}{c} F^{\alpha\beta} U_\alpha U_\beta = 0,$$

which is generally not true. Consider a medium with charge density at rest,  $U_\alpha J^\alpha = c^2 \rho$ . Therefore, we need an extra term to cancel this one. It is clear that

$$\frac{1}{c^2} (U_\beta J^\beta) U^\alpha$$

will do. Then, we will arrive at

$$J^\alpha - \frac{1}{c^2} (U_\beta J^\beta) U^\alpha = \frac{\sigma}{c} F^{\alpha\beta} U_\beta.$$

(b) Since  $(c\rho, \mathbf{J})$  is a 4-vector, we know it transforms as

$$\mathbf{J} = \mathbf{J}' + \frac{\gamma - 1}{\beta^2} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{J}') + \gamma c \rho' \boldsymbol{\beta}, \quad c\rho = \gamma (c\rho' + \boldsymbol{\beta} \cdot \mathbf{J}'),$$

where we can write the second transform as

$$\gamma c \rho' = c\rho - \gamma \boldsymbol{\beta} \cdot \mathbf{J}'.$$

Therefore,

$$\mathbf{J} = \mathbf{J}' + \frac{\gamma - 1}{\beta^2} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{J}') + c\rho \boldsymbol{\beta} - \gamma \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{J}') = \mathbf{J}' + \left[ \frac{\gamma - 1}{\beta^2} - \gamma \right] \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{J}') + \rho \mathbf{v}.$$

Now, in the rest frame of the medium,  $\mathbf{J}' = \sigma \mathbf{E}'$ , and  $\mathbf{E}'$  is related to its strength in the laboratory frame by

$$\mathbf{E}' = \gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}),$$

we have

$$\begin{aligned} \mathbf{J} &= \sigma \left[ \gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right. \\ &\quad \left. + \left( \frac{\gamma - 1}{\beta^2} - \gamma \right) \boldsymbol{\beta} \left\{ \boldsymbol{\beta} \cdot \left[ \gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right] \right\} \right] + \rho \mathbf{v} \\ &= \sigma \left[ \gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) + \left( \frac{\gamma - 1}{\beta^2} - \gamma \right) \left( \gamma - \frac{\beta^2 \gamma^2}{\gamma + 1} \right) \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right] + \rho \mathbf{v}. \end{aligned}$$

Since  $\beta^2 = 1 - \gamma^{-2}$ , then

$$\gamma - \frac{\beta^2 \gamma^2}{\gamma + 1} = \gamma - \frac{\gamma^2 - 1}{\gamma + 1} = 1,$$

and

$$\mathbf{J} = \sigma \left[ \gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) + \left( \frac{\gamma - 1}{\beta^2} - \frac{\gamma^2}{\gamma + 1} - \gamma \right) \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right] + \rho \mathbf{v}.$$

Also,

$$\frac{\gamma - 1}{\beta^2} = \frac{\gamma^2}{\gamma + 1},$$

we will finally have

$$\mathbf{J} = \gamma \sigma [\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B} - \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E})] + \rho \mathbf{v}.$$

(c) If the medium is initially uncharged, then  $U_\alpha J^\alpha = 0$ , and the equation becomes

$$J^\alpha = \frac{\sigma}{c} F^{\alpha\beta} U_\beta.$$

For the spatial component, we have

$$J^i = \frac{\sigma}{c} (F^{i0} U^0 - F^{ij} U^j) = \frac{\sigma}{c} (E_i \gamma c - (-\epsilon_{ijk} B_k) \gamma v_j) = \gamma \sigma (E_i + (\mathbf{v} \times \mathbf{B})_i),$$

which leads to  $\mathbf{J} = \gamma \sigma [\mathbf{E} + \mathbf{v} \times \mathbf{B}]$ . In the classical limit,  $\gamma \rightarrow 1$ , this reduces to the known result.